

Multipermutation Ulam Sphere Analysis Toward Characterizing Maximal Code Size

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Abstract—Permutation codes, in the form of *rank modulation*, have shown promise for applications such as flash memory. One of the metrics recently suggested as appropriate for rank modulation is the Ulam metric. Multipermutation codes have also been proposed as a generalization of permutation codes that would improve code size. In this paper we analyze the Ulam metric in the context of multipermutations, noting similarities and differences with the Ulam metric in the context of permutations. We then consider sphere sizes for multipermutations under the Ulam metric and resulting bounds on code size.

I. INTRODUCTION

Permutation (and multipermutation) codes were invented as early as the 1960's, when Slepian proposed constructing a code by permuting the order of the numbers of an initial sequence [13]. More recently, Jiang et al. proposed permutation codes utilizing the Kendall- τ metric for use in flash memory via the *rank modulation* scheme [8]. Since then, permutation codes and their generalization to multipermutation codes have been a hot topic in the research community with various related schemes being suggested [1], [2], [3], [4], [9], [11].

One scheme of particular interest was the proposal of Farnoud et al. to utilize the Ulam metric in place of the Kendall- τ metric [3]. Errors in flash memory devices occur when cell charges leak or when rewriting operations cause overshoot errors resulting in inaccurate charge levels. While the Kendall- τ metric is suitable for correcting relatively small errors of this nature, the Ulam metric would be more robust to large charge leakages or overshoot errors within a cell.

However, there is a trade-off in code size when rank modulation is used in conjunction with the Ulam metric instead of the Kendall- τ metric. The Ulam distance between permutations is always less than or equal to the Kendall- τ distance between permutations, which implies that the maximum code size for a permutation code utilizing the Ulam metric is less than or equal to the maximum code size of a permutation code utilizing the Kendall- τ metric [3]. One possible compensation for this trade-off is the generalization from permutation codes to multipermutation codes.

In flash memory devices, permutations or multipermutations may be modeled physically by relative cell charge rankings. The number of possible messages is limited by the number of distinguishable relative rankings. However, it was shown in [4] that multipermutations may significantly increase the total possible messages compared to ordinary permutations.

For example, if only k different charge levels are possible, then permutations of length k can be stored. Hence, in r blocks of length k , one may store $(k!)^r$ potential messages. On the other hand, if one uses multipermutations in the same set of blocks, then $(kr)!/(r!)^k$ potential messages are possible.

Bounds on permutation codes in the Ulam metric were studied in [3] and [7]. In [10], the nonexistence of nontrivial perfect permutation codes in the Ulam metric was proven by examining the size of Ulam spheres, spheres comprised of all permutations within a given Ulam distance of a particular permutation. However, no similar study of Multipermutation Ulam spheres exists, and currently known bounds on code size do not always consider the problem of differing sphere sizes. The current paper examines Ulam sphere sizes in the context of multipermutations and provides new bounds on code size.

The paper is organized as follows: First, Section II defines notation and basic concepts used in the paper. Next, Section III compares properties of the Ulam metric as defined for permutations and multipermutations, and then provides a simplification of the r -regular Ulam metric for multipermutations (Lemmas 1 and 2). Section IV considers an application of Young Tableaux and the RSK-correspondence to calculate r -regular Ulam sphere sizes (Lemma 5 and Prop. 6). Section V then discusses duplicate translocation sets and a method of calculating the size of spheres of radius $t = 1$ for any center (Thm. 12). Section VI follows, demonstrating minimal and maximal sphere sizes (Lemmas 13 and 15) and providing both lower and upper bounds on code size (Lemmas 14, 16, and 17). Proofs are omitted because of space constraints, but are available in the full version of this paper [15].

II. PRELIMINARIES AND NOTATION

In this section we introduce notation and definitions used in this paper. Unless otherwise stated, definitions are based on conventions established in [3], [4], [10], and [11]. Throughout this paper n and r are assumed to be positive integers, r dividing n . The notation $[n]$ denotes the set $\{1, 2, \dots, n\}$ and $\mathbb{S}_n$ denotes the set of permutations on $[n]$, i.e. the symmetric group of size $n!$. For $\sigma \in \mathbb{S}_n$, we write $\sigma = [\sigma(1), \sigma(2), \dots, \sigma(n)]$, where for all $i \in [n]$, $\sigma(i)$ is the image of i under σ . Throughout this paper we assume $\sigma, \pi \in \mathbb{S}_n$. With a slight abuse of notation, we may also use σ to mean the sequence $(\sigma(1), \sigma(2), \dots, \sigma(n)) \in \mathbb{Z}^n$ associated with $\sigma \in \mathbb{S}_n$. Multi-

plication of permutations is defined by composition so that for all $i \in [n]$, we have $(\sigma\tau)(i) = \sigma(\tau(i))$.

An r -regular multiset is a multiset such that each of its elements is repeated r times. A *multi-permutation* is an ordered tuple of the elements of a multiset, and in the instance of an r -regular multiset, is called an r -regular *multi-permutation*. Following the work of [4] and [11], this study focuses on r -regular multi-permutations.

For each $\sigma \in \mathbb{S}_n$ we define a corresponding r -regular multi-permutation $\mathbf{m}_\sigma^r$ as follows: for all $i \in [n]$ and $j \in [n/r]$,

$$\mathbf{m}_\sigma^r(i) := j \text{ if and only if } (j-1)r+1 \leq \sigma(i) \leq jr,$$

and $\mathbf{m}_\sigma^r := (\mathbf{m}_\sigma^r(1), \mathbf{m}_\sigma^r(2), \dots, \mathbf{m}_\sigma^r(n)) \in \mathbb{Z}^n$. For example, if $n = 6$, $r = 2$, and $\sigma = [1, 5, 2, 4, 3, 6]$, then $\mathbf{m}_\sigma^r = (1, 3, 1, 2, 2, 3)$. This definition follows the convention of [11] rather than that of [4], so that properties of the Ulam metric for permutations will carry over to the Ulam metric for multi-permutations (Lemmas 1 and 2 of Section III). We say that $\sigma \equiv_r \pi$ if and only if $\mathbf{m}_\sigma^r = \mathbf{m}_\pi^r$. The equivalence class $R_r(\sigma)$ of $\sigma \in \mathbb{S}_n$ is defined as $R_r(\sigma) := \{\pi \in \mathbb{S}_n : \pi \equiv_r \sigma\}$. For a subset $S \subseteq \mathbb{S}_n$, the notation $\mathcal{M}_r(S) := \{\mathbf{m}_\sigma^r : \sigma \in S\}$, i.e. the set of r -regular multi-permutations corresponding to elements of S .

The following definition is our own. For any $\mathbf{m} \in \mathbb{Z}^n$, and $\sigma \in \mathbb{S}_n$, we define the product (a right group action) $\mathbf{m} \cdot \sigma$ by composition, similarly to the definition of multiplication of permutations. More precisely, for all $i \in [n]$, let $(\mathbf{m} \cdot \sigma)(i) := \mathbf{m}(\sigma(i))$. With this definition, notice that $\mathbf{m}_\sigma^r \cdot \pi = \mathbf{m}_{\sigma\pi}^r$. It is possible for different permutations to correspond to the same multi-permutation, but for any $\tau \in \mathbb{S}_n$, it is clear that $\mathbf{m}_\sigma^r = \mathbf{m}_\pi^r$ implies $\mathbf{m}_\sigma^r \cdot \tau = \mathbf{m}_\pi^r \cdot \tau$.

We finish this section by defining what a multi-permutation code is. A subset $C \subseteq \mathbb{S}_n$ is called an r -regular *multi-permutation code* if and only if for all $\sigma \in C$, we also have $R_r(\sigma) \subseteq C$. Such a code is denoted by $\text{MPC}(n, r)$, and we say that C is an $\text{MPC}(n, r)$. If C is an $\text{MPC}(n, r)$, whenever a permutation is a member of C its entire equivalence class is also contained within C , so the code can be represented by the set of r -regular multi-permutations associated with elements of C , i.e. the set $\mathcal{M}_r(C)$. Moreover, if C is an $\text{MPC}(n, r)$, we define the cardinality $|C|_r$ as $|C|_r := |\mathcal{M}_r(C)|$ (this notation and definition differs slightly from [4]).

III. MULTIPER-MUTATION ULAM METRIC

In this section we discuss some similarities and differences between the Ulam metric for permutations and the Ulam metric for multi-permutations. We begin by defining the Ulam metric for permutations.

For any two sequences $\mathbf{u}, \mathbf{v} \in \mathbb{Z}^n$, $\ell(\mathbf{u}, \mathbf{v})$ denotes the length of the longest common subsequence of $\mathbf{u}$ and $\mathbf{v}$. In other words, $\ell(\mathbf{u}, \mathbf{v})$ is the largest integer $k \in \mathbb{Z}_{>0}$ such that there exists a sequence $(a_1, a_2, \dots, a_k)$ where for all $p \in [k]$, we have $a_p = \mathbf{u}(i_p) = \mathbf{v}(j_p)$ with $1 \leq i_1 < i_2 < \dots < i_k \leq n$ and $1 \leq j_1 < j_2 < \dots < j_k \leq n$. The *Ulam distance* $d_o(\sigma, \pi)$ between permutations $\sigma, \pi \in \mathbb{S}_n$ is defined as $d_o(\sigma, \pi) := n - \ell(\sigma, \pi)$.

It is also known that the Ulam distance $d_o(\sigma, \pi)$ between $\sigma, \pi \in \mathbb{S}_n$ is equivalent to the minimum number of translocations needed to transform σ into π [3]. Here, for distinct $i, j \in [n]$, the translocation $\phi(i, j) \in \mathbb{S}_n$ is defined as follows:

$$\phi(i, j) := \begin{cases} [1, 2, \dots, i-1, i+1, i+2, \dots, j, i, j+1, \dots, n] & \text{if } i < j \\ [1, 2, \dots, j-1, i, j, j+1, \dots, i-1, i+1, \dots, n] & \text{otherwise} \end{cases}$$

The notation $\phi(i, i)$ is understood to mean the identity permutation $[1, 2, \dots, n] \in \mathbb{S}_n$, denoted by e . When it is not necessary to specify any index, a translocation may be written simply as ϕ . Intuitively, when multiplied on the right of a permutation $\sigma \in \mathbb{S}_n$, the translocation $\phi(i, j) \in \mathbb{S}_n$ deletes $\sigma(i)$ from the i th position of σ and then inserts it in the new j th position (shifting positions between i and j in the process).

The r -regular *Ulam distance* $d_o^r(\sigma, \pi)$ between permutations $\sigma, \pi \in \mathbb{S}_n$ is defined as the minimum Ulam distance among all members of $R_r(\sigma)$ and $R_r(\pi)$. That is, $d_o^r(\sigma, \pi) := \min_{\sigma' \in R_r(\sigma), \pi' \in R_r(\pi)} d_o(\sigma', \pi')$. Notice that the r -regular Ulam distance is defined over equivalence classes.

Although technically a distance between equivalence classes, it is convenient to think of the r -regular Ulam distance instead as a distance between multi-permutations. Viewed this way, the property of the Ulam metric for permutations, that it can be defined in terms of longest common subsequences or equivalently in terms of translocations, carries over to the r -regular Ulam distance. The next lemma states that the r -regular Ulam distance between permutations σ and π is equal to n minus the length of the longest common subsequence of their corresponding r -regular multi-permutations.

Lemma 1. $d_o^r(\sigma, \pi) = n - \ell(\mathbf{m}_\sigma^r, \mathbf{m}_\pi^r)$.

If two multi-permutations $\mathbf{m}_\sigma^r$ and $\mathbf{m}_\pi^r$ have a common subsequence of length k , then $\mathbf{m}_\sigma^r$ can be transformed into $\mathbf{m}_\pi^r$ with $n-k$ (but no fewer) delete/insert operations. Delete/insert operations correspond to applying (multiplying on the right) a translocation. Hence by Lemma 1 we can state the following lemma about the r -regular Ulam distance.

Lemma 2. $d_o^r(\sigma, \pi) = \min\{k \in \mathbb{Z}_{>0} : \text{there exists } (\phi_1, \phi_2, \dots, \phi_k) \text{ s.t. } \mathbf{m}_\sigma^r \cdot \phi_1 \phi_2 \dots \phi_k = \mathbf{m}_\pi^r\}$.

Lemmas 1 and 2 allow us to view the Ulam metric for r -regular multi-permutations similarly to the way we view the Ulam metric for permutations; in terms of longest common subsequences or in terms of the minimum number of translocations. Another known property of the Ulam metric for permutations is left invariance, i.e. given $\tau \in \mathbb{S}_n$, we have $d_o(\sigma, \pi) = d_o(\tau\sigma, \tau\pi)$. However, left invariance does not hold in general for multi-permutations.

Lemma 3. Let $n/r \geq 2$ and $r \geq 2$. Then there exist $\sigma', \pi' \in \mathbb{S}_n$ such that $d_o^r(e, \sigma') \neq d_o^r(\pi' e, \pi' \sigma')$.

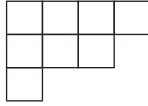
The fact that left invariance does not hold for the r -regular Ulam metric has implications on r -regular Ulam sphere sizes, defined and discussed in the next section. Left invariance im-

plies sphere size does not depend upon the center. However, we will demonstrate that in the multipermutation case sizes may differ depending upon the center, a fact previously unknown.

IV. YOUNG TABLEAUX SPHERE SIZE CALCULATION

In [10], Young tableaux and the RSK-Correspondence were utilized to calculate Ulam Sphere sizes. A similar approach can be applied to r -regular Ulam spheres of arbitrary radius centered at $\mathbf{m}_e^r$. It is first necessary to introduce some basic notation and definitions regarding Young tableaux. Additional information on the subject can be found in [6], [13], and [14].

A *Young diagram* is a left-justified collection of cells with a (weakly) decreasing number of cells in each row below. Listing the number of cells in each row gives a partition $\lambda = (\lambda_1, \lambda_2, \dots, \lambda_k)$ of n , where n is the total number of cells in the Young diagram. The notation $\lambda \vdash n$ indicates λ is a partition of n . Because the partition $\lambda \vdash n$ defines a unique Young diagram and vice versa, a Young diagram may be referred to by its associated partition $\lambda \vdash n$. For example, the partition $\lambda := (4, 3, 1) \vdash 8$ has the corresponding Young diagram pictured below.



A *Young tableau*, or simply a *tableau*, is a filling of a Young diagram $\lambda \vdash n$ such that values in all cells are weakly increasing across each row and strictly increasing down each column. If each of the integers 1 through n appears exactly once in a tableau T that is a filling of a Young diagram $\lambda \vdash n$, then we call T a *standard Young tableau*, abbreviated *SYT*.

The *RSK-correspondence* ([6], [14]) provides a bijection between r -regular multipermutations $\mathbf{m}_\sigma^r$ and ordered pairs (P, Q) on the same Young diagram $\lambda \vdash n$, where P is a tableaux whose members come from $\mathbf{m}_\sigma^r$ and Q is a *SYT*. A stronger form of the following lemma appears in [6].

Lemma 4. *Let $\sigma \in S_n$ and P be the tableau resulting from running the Schensted algorithm on the entries of σ . Then the number of columns in P is equal to $\ell(\mathbf{m}_e^r, \mathbf{m}_\sigma^r)$, the length of the longest non-decreasing subsequence of $\mathbf{m}_\sigma^r$.*

The above lemma, in conjunction with the RSK-correspondence, means that for all $k \in [n]$, the size of the set $\{\mathbf{m}_\sigma^r \in \mathcal{M}_r(\mathbb{S}_n) : \ell(\mathbf{m}_e^r, \mathbf{m}_\sigma^r) = k\}$ is equal to the sum of the number of ordered pairs (P, Q) on each Young diagram $\lambda \vdash n$ such that $\lambda_1 = k$, where P is a tableaux whose members come from $\mathbf{m}_\sigma^r$ and Q is a *SYT*. The number of *SYT* on a particular $\lambda \vdash n$ is denoted by f^λ . We denote by K_r^λ (our own notation) the number of Young tableaux on $\lambda \vdash n$ such that each $i \in [n/r]$ appears exactly r times.

In the case of permutations, recall that the Ulam sphere $S(\sigma, t)$ centered at σ of radius t was defined as $S(\sigma, t) := \{\pi \in \mathbb{S}_n : d_o(\sigma, \pi) \leq t\}$, which is equivalent by definition to the set $\{\pi \in \mathbb{S}_n : n - \ell(\sigma, \pi) \leq t\}$. In the case of r -regular multipermutations, for $t \in \mathbb{Z}_{>0}$, we introduce the following analogous definition of a sphere. Define $S(\mathbf{m}_\sigma^r, t) := \{\mathbf{m}_\pi^r \in \mathcal{M}_r(\mathbb{S}_n) : d_o^r(\sigma, \pi) \leq t\}$. We call $S(\mathbf{m}_\sigma^r, t)$ the *r -regular*

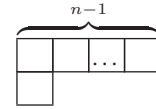
Ulam sphere centered at $\mathbf{m}_\sigma^r$ of radius t . By Lemma 1, $S(\mathbf{m}_\sigma^r, t) = \{\mathbf{m}_\pi^r \in \mathcal{M}_r(\mathbb{S}_n) : n - \ell(\mathbf{m}_\sigma^r, \mathbf{m}_\pi^r) \leq t\}$. It should be noted, however, that the notation $\mathbf{m}_\pi^r$ is a bit misleading in the sphere definition because given $\mathbf{m}_\pi^r \in \mathcal{M}(\mathbb{S}_n)$, we cannot uniquely determine π . The next lemma states the relationship between f^λ , K_r^λ , and $|S(\mathbf{m}_e^r, t)|$.

Lemma 5. *Let $t \in [0, n-1]$, and $\Lambda := \{\lambda \vdash n : \lambda_1 \geq n-t\}$. Then $|S(\mathbf{m}_e^r, t)| = \sum_{\lambda \in \Lambda} (f^\lambda)(K_r^\lambda)$.*

The following proposition is an application of Lemma 5.

Proposition 6. $|S(\mathbf{m}_e^r, 1)| = 1 + (n-1)(n/r - 1)$.

Proof. First note that $|S(\mathbf{m}_e^r, 0)| = 1$. There is only one possible partition $\lambda \vdash n$ such that $\lambda_1 = n-1$, namely $\lambda' := (n-1, 1)$, with its Young diagram pictured below.



Therefore by Lemma 5, $|S(\mathbf{m}_e^r, 1)| = 1 + (f^{\lambda'})(K_r^{\lambda'})$. Applying the well-known hook length formula ([5], [6]), we obtain $f^{\lambda'} = n-1$. The value $K_r^{\lambda'}$ is characterized by possible fillings of row 2 with the stipulation that each $i \in [n/r]$ must appear exactly r times in the diagram. In this case, since there is only a single box in row 2, the possible fillings are $i \in [n/r - 1]$, each of which yields a unique Young tableau of the desired type. Hence $K_r^{\lambda'} = n/r - 1$, which implies that $|S(\mathbf{m}_e^r, 1)| = 1 + (n-1)(n/r - 1)$. $\square$

Proposition 6 demonstrates how Young Tableaux may be used to calculate r -regular Ulam spheres centered at $\mathbf{m}_e^r$.

V. r -REGULAR ULAM SPHERES AND DUPLICATION SETS

In the previous section we showed how multipermutation Ulam spheres may be calculated when the center is $\mathbf{m}_e^r$. In this section we provide a way to calculate sphere sizes for any center when the radius is $t = 1$. The r -regular Ulam sphere sizes play an important role in understanding the potential code sizes for a given minimum distance. For example, the well-known sphere-packing and Gilbert-Varshamov bounds rely on calculating, or at least bounding sphere sizes.

Lemma 5 provided a way to calculate r -regular Ulam spheres centered at $\mathbf{m}_e^r$. Unfortunately, the choice of center has an impact on the size of the sphere, as is easily confirmed by comparing Proposition 6 to Lemma 15 (Section VI). Hence the applicability of Lemma 5 is limited.

We begin to address the issue of differing sphere sizes by first considering the radius $t = 1$ case. By Lemma 2, the r -regular Ulam sphere definition can be viewed in terms of translocations. That is, $S(\mathbf{m}_\sigma^r, t)$ is equivalent to $\{\mathbf{m}_\pi^r \in \mathcal{M}_r(\mathbb{S}_n) : \text{there exists } k \in [t] \text{ and } (\phi_1, \dots, \phi_k) \text{ s.t. } \mathbf{m}_\sigma^r \cdot \phi_1 \cdots \phi_k = \mathbf{m}_\pi^r\}$. To aid with calculating sphere sizes, we introduce (as our own definition) the following subset of the set of translocations. Define

$$T_n := \{\phi(i, j) \in \mathbb{S}_n : i - j \neq 1\}.$$

We call T_n the **unique set of translocations**. By definition, T_n is the set of all translocations in $\mathbb{S}_n$, except translocations of the form $\phi(i, i-1)$. We exclude translocations of this form because they can be modeled by translocations of the form $\phi(i-1, i)$, and are therefore redundant.

We claim that the set T_n is precisely the set of translocations needed to obtain all unique permutations within the Ulam sphere of radius 1 via multiplication. Moreover, there is no redundancy in the set, that is, there is no smaller set of translocations yielding the entire Ulam sphere of radius 1 when multiplied with a given center permutation. These facts are stated in the next lemma.

Lemma 7. $S(\sigma, 1) = \{\sigma\phi \in \mathbb{S}_n : \phi \in T_n\}$ and $|T_n| = |S(\sigma, 1)|$.

In the case of permutations, the set T_n has no redundancies. If $\phi_1, \phi_2 \in T_n$, then $\sigma\phi_1 = \sigma\phi_2$ implies $\phi_1 = \phi_2$. Alternatively, in the case of multipermutations, the set T_n can generally be shrunk to exclude redundancies. Notice that $S(\mathbf{m}_\sigma^r, 1) = \{\mathbf{m}_\sigma^r \in \mathcal{M}_r(\mathbb{S}_n) : \text{there exists } \phi \text{ s.t. } \mathbf{m}_\sigma^r \cdot \phi = \mathbf{m}_\sigma^r\}$, which is equal to $\{\mathbf{m}_\sigma^r \cdot \phi \in \mathcal{M}_r(\mathbb{S}_n) : \phi \in T_n\}$. However, it is possible that there exist $\phi_1, \phi_2 \in T_n$ such that $\phi_1 \neq \phi_2$, but $\mathbf{m}_\sigma^r \cdot \phi_1 = \mathbf{m}_\sigma^r \cdot \phi_2$. In such an instance we may refer to either ϕ_1 or ϕ_2 as a **duplicate translocation** for $\mathbf{m}_\sigma^r$.

If we remove all duplicate translocations for $\mathbf{m}_\sigma^r$ from T_n , then the resulting set will have the same cardinality as the r -regular Ulam sphere of radius 1 centered at $\mathbf{m}_\sigma^r$. The next definition (our own) is the set of standard duplicate translocations. For the remainder of the paper, assume that $\mathbf{m}$ is an n -length integer tuple, i.e. $\mathbf{m} \in \mathbb{Z}^n$. Define

$$D(\mathbf{m}) := \{ \phi(i, j) \in T_n \setminus \{e\} : (\mathbf{m}(i) = \mathbf{m}(j)) \text{ or } (\mathbf{m}(i) = \mathbf{m}(i-1)) \}$$

Call $D(\mathbf{m})$ the **standard duplicate translocation set** for $\mathbf{m}$.

If we take an r -regular multipermutation $\mathbf{m}_\sigma^r$, then removing $D(\mathbf{m}_\sigma^r)$ from T_n equates to removing a set of duplicate translocations. These duplications come in two varieties. The first variety corresponds to the first condition of the $D(\mathbf{m})$ definition, when $\mathbf{m}(i) = \mathbf{m}(j)$. For example, if $\sigma \in \mathbb{S}_6$ such that $\mathbf{m}_\sigma^2 = (1, 3, 2, 2, 3, 1)$, then we have $\mathbf{m}_\sigma^2 \cdot \phi(1, 5) = (3, 2, 2, 3, 1, 1) = \mathbf{m}_\sigma^2 \cdot \phi(1, 6)$, since $\mathbf{m}_\sigma^2(2) = 3 = \mathbf{m}_\sigma^2(4)$. This is because moving the first 1 to the left or to the right of the last 1 results in the same tuple.

The second variety corresponds to the second condition of the $D(\mathbf{m})$ definition above, when $\mathbf{m}(i) = \mathbf{m}(i-1)$. For example, if $\mathbf{m}_\sigma^2 = (1, 3, 2, 2, 3, 1)$ as before, then for all $j \in \{1, 2, 3, 4, 5, 6\}$, we have $\mathbf{m}_\sigma^2 \cdot \phi(3, j) = \mathbf{m}_\sigma^2 \cdot \phi(4, j)$. This is because any translocation that deletes and inserts the second of the two adjacent 2's does not result in a different tuple when compared to deleting and inserting the first of the two adjacent 2's.

Lemma 8. $S(\mathbf{m}_\sigma^r, 1) = \{\mathbf{m}_\sigma^r \cdot \phi \in \mathcal{M}_r(\mathbb{S}_n) : \phi \in T_n \setminus D(\mathbf{m}_\sigma^r)\}$.

While Lemma 8 indicates that $D(\mathbf{m}_\sigma^r)$ is a set of duplicate translocations for $\mathbf{m}_\sigma^r$, we have not shown that $T_n \setminus D(\mathbf{m}_\sigma^r)$ is the set of minimal size having the quality that $S(\mathbf{m}_\sigma^r, 1) =$

$\{\mathbf{m}_\sigma^r \cdot \phi \in \mathcal{M}_r(\mathbb{S}_n) : \phi \in T_n \setminus D(\mathbf{m}_\sigma^r)\}$. In fact it is not minimal. In some instances it is possible to remove further duplicate translocations to reduce the set size. We will define another set of duplicate translocations, but a few preliminary definitions are first necessary.

We say that $\mathbf{m}$ is *alternating* if for all odd integers $1 \leq i \leq n$, $\mathbf{m}(i) = \mathbf{m}(1)$ and for all even integers $2 \leq i' \leq n$, $\mathbf{m}(i') = \mathbf{m}(2)$ but $\mathbf{m}(1) \neq \mathbf{m}(2)$. In other words, any alternating tuple is of the form $(a, b, a, b, \dots, a, b)$ or $(a, b, a, b, \dots, a)$ where $a, b \in \mathbb{Z}$ and $a \neq b$. Any singleton is also said to be alternating. Now for integers $1 \leq i \leq n$ and $0 \leq k \leq n-i$, the *substring* $\mathbf{m}[i, i+k]$ of $\mathbf{m}$ is defined as $\mathbf{m}[i, i+k] := (\mathbf{m}(i), \mathbf{m}(i+1), \dots, \mathbf{m}(i+k))$. Given a substring $\mathbf{m}[i, j]$ of $\mathbf{m}$, the *length* $|\mathbf{m}[i, j]|$ of $\mathbf{m}[i, j]$ is defined as $|\mathbf{m}[i, j]| := j - i + 1$. As an example, if $\mathbf{m}' := (1, 2, 2, 4, 2, 4, 3, 1, 3)$, then $\mathbf{m}'[3, 6] = (2, 4, 2, 4)$ is an alternating substring of $\mathbf{m}'$ of length 4.

Next define

$$E(\mathbf{m}) := \{ \begin{array}{l} \phi(i, j) \in T_n \setminus D(\mathbf{m}) : \\ i < j \text{ and there exists } k \in [i, j-2] \text{ s.t.} \\ (\phi(j, k) \in T_n \setminus D(\mathbf{m})) \\ \text{and } (\mathbf{m} \cdot \phi(i, j) = \mathbf{m} \cdot \phi(j, k)) \end{array} \}$$

We call $E(\mathbf{m})$ the **alternating duplicate translocation set** for $\mathbf{m}$ because it is only nonempty when $\mathbf{m}$ contains an alternating substring of length at least 4. To calculate $|E(\mathbf{m}_\sigma^r)|$, we define a set of equal size that is easier to calculate. Let $E^*(\mathbf{m}) := \{(i, j) \in [n] \times [n] : (\mathbf{m}[i, j] \text{ is alternating and } (|\mathbf{m}[i, j]| \geq 4) \text{ and } (|\mathbf{m}[i, j]| \text{ is even}))\}$.

Lemma 9. $|E(\mathbf{m})| = |E^*(\mathbf{m})|$

Calculating $|E^*(\mathbf{m})|$ equates to counting the number of alternating substrings of $\mathbf{m}$ whose length is even and at least 4. The following lemma helps to simplify this calculation further.

Lemma 10. Let $n \geq 4$ and let $\mathbf{m}$ be alternating. Then

$$\begin{array}{l} 1) \text{ If } n \text{ is even then } |E^*(\mathbf{m})| = \left(\frac{n-2}{2}\right)^2 \\ 2) \text{ If } n \text{ is odd then } |E^*(\mathbf{m})| = \left(\frac{n-3}{2}\right) \left(\frac{n-1}{2}\right) \end{array}$$

Notice that by Lemma 10, it suffices to calculate $|E(\mathbf{m})|$ for locally maximal length alternating substrings of $\mathbf{m}$. An alternating substring $\mathbf{m}[i, j]$ of $\mathbf{m}$ is of *locally maximal length* if and only if 1) $\mathbf{m}[i-1]$ is not alternating or $i = 1$; and 2) $\mathbf{m}[i, j+1]$ is not alternating or $j = n$.

Lemma 11. Let $\phi_1, \phi_2 \in T_n \setminus (D(\mathbf{m}_\sigma^r) \cup E(\mathbf{m}_\sigma^r))$. Then $\phi_1 = \phi_2$ if and only if $\mathbf{m}_\sigma^r \cdot \phi_1 = \mathbf{m}_\sigma^r \cdot \phi_2$.

Lemma 11 implies that we can calculate r -regular Ulam sphere sizes of radius 1 whenever we can calculate the appropriate duplication set. This calculation can be simplified by noting that $D(\mathbf{m}) \cap E(\mathbf{m}) = \emptyset$ (by the definition of $E(\mathbf{m})$) and then decomposing the duplication set into these components. This idea is stated in the next theorem.

Theorem 12. $|S(\mathbf{m}_\sigma^r, 1)| = 1 + (n-1)^2 - |D(\mathbf{m}_\sigma^r)| - |E(\mathbf{m}_\sigma^r)|$.

Theorem 12 reduces the calculation of $|S(\mathbf{m}_\sigma^r, 1)|$ to calculating $|D(\mathbf{m}_\sigma^r)|$ and $|E(\mathbf{m}_\sigma^r)|$. It is an easy matter to calculate $|D(\mathbf{m}_\sigma^r)|$, since it is exactly equal to $(n-2)$ times the number of $i \in [n]$ such that $i = 1$ or $\mathbf{m}_\sigma^r(i) = \mathbf{m}_\sigma^r(i-1)$ plus $(r-1)$ times the number of $i \in [n]$ such that $\mathbf{m}_\sigma^r(i) \neq \mathbf{m}_\sigma^r(i-1)$. We also showed how to calculate $|E(\mathbf{m})|$ earlier. The next example is an application of Theorem 12

Example. Suppose $\sigma := [1, 2, 3, 4, 9, 6, 7, 11, 5, 10, 12, 8]$. Then $\mathbf{m}_\sigma^3 := (1, 1, 1, 2, 3, 2, 3, 2, 4, 4, 3, 4)$. There are 3 values of $i \in [12]$ such that $\mathbf{m}_\sigma^3(i) = \mathbf{m}_\sigma^3(i-1)$, which implies that $|D(\mathbf{m}_\sigma^3)| = (3)(12-2) + (12-3)(3-1) = 48$. Meanwhile, by Lemmas 9 and 10, $|E(\mathbf{m}_\sigma^3)| = ((5-3)/2)((5-1)/2) = 2$. By Theorem 12, $|S(\mathbf{m}_\sigma^3, 1)| = (12-1)^2 - 48 - 2 = 71$.

VI. MIN/MAX SPHERES AND CODE SIZE BOUNDS

In this section we show choices of center achieving minimum and maximum r -regular Ulam sphere sizes for the radius $t = 1$ case. The minimum and maximum values are explicitly given. We then discuss resulting bounds on code size. First let us consider the r -regular Ulam sphere of minimal size.

Lemma 13. $|S(\mathbf{m}_e^r, 1)| \leq |S(\mathbf{m}_\sigma^r, 1)|$

Lemma 13, along with Proposition 6 implies that the r -regular Ulam sphere size of radius $t = 1$ is bounded (tightly) below by $(1 + (n-1)(n/r - 1))$. This in turn implies the following sphere-packing type upper bound on any single error-correcting code.

Lemma 14. Let C be a single-error correcting $\text{MPC}_o(n, r)$. Then

$$|C|_r \leq \frac{n!}{(r!)^{n/r} (1 + (n-1)(n/r - 1))}.$$

We have seen that $|S(\mathbf{m}_\sigma^r)|$ is minimized when $\sigma = e$. We now discuss the choice of center yielding maximal sphere size. Define $\omega \in \mathbb{S}_n$ as follows: $\omega(i) := ((i-1) \bmod (n/r))r + \lceil ir/n \rceil$ and $\omega := [\omega(1), \omega(2), \dots, \omega(n)]$. With this definition, for all $i \in [n]$, we have $\mathbf{m}_\omega^r(i) = i \bmod (n/r)$. For example, if $r = 3$ and $n = 12$, then $\omega = [1, 4, 7, 10, 2, 5, 8, 11, 3, 6, 9, 12]$ and $\mathbf{m}_\omega^r = (1, 2, 3, 4, 1, 2, 3, 4, 1, 2, 3, 4)$. We can use Theorem 12 to calculate $|S(\mathbf{m}_\omega^r, 1)|$, and then show that this is the maximal r -regular Ulam sphere size (except for the case when $n/r = 2$).

Lemma 15. Let $n/r \neq 2$. Then $|S(\mathbf{m}_\sigma^r, 1)| \leq |S(\mathbf{m}_\omega^r, 1)|$ and if $n/r > 2$, then $|S(\mathbf{m}_\omega^r, 1)| = (1 + (n-1)^2) - (r-1)n$.

Extending the concept of perfect permutation codes discussed in [10], we define a perfect multipermutation code. Let C be an $\text{MPC}(n, r)$. Then C is a perfect t -error correcting code if and only if for all $\sigma \in \mathbb{S}_n$, there exists a unique $\mathbf{m}_\sigma^r \in M_r(C)$ such that $\mathbf{m}_\sigma^r \in S(\mathbf{m}_\sigma^r, t)$. We call such C a **perfect t -error correcting $\text{MPC}(n, r)$** . With this definition the upper bound of Lemma 15 implies a lower bound on a perfect single-error correcting $\text{MPC}(n, r)$.

Lemma 16. Let C be a perfect single-error correcting $\text{MPC}(n, r)$ and $n/r \neq 2$. Then

$$|C|_r \geq \frac{n!}{(r!)^{n/r} ((1 + (n-1)^2) - (r-1)n)}.$$

A more general lower bound is easily obtained by applying Lemma 15 with a standard Gilbert-Varshamov bound argument. In the lemma statement, C is an $\text{MPC}_o(n, r, d)$ if and only if C is an $\text{MPC}(n, r)$ such that $\min_{\sigma, \pi \in C, \sigma \neq \pi} d_o^r(\sigma, \pi) = d$. While the lower bound of Lemma 16 applies only to perfect codes that are $\text{MPC}(n, r, d)$ with $d \geq 3$, the next lemma applies to any $\text{MPC}(n, r, d)$, which may not be perfect.

Lemma 17. Let $n/r \neq 2$ and C be an $\text{MPC}_o(n, r, d)$ of maximal cardinality. Then

$$|C|_r \geq \frac{n!}{(r!)^{n/r} (1 + (n-1)^2 - (r-1)n)^{d-1}}.$$

VII. CONCLUSION

This paper compared the Ulam metric for the permutation and multipermutation cases, providing a simplification of the r -regular Ulam metric. The fact that r -regular Ulam sphere sizes differ depending upon the center was also shown. New methods for calculating the size of r -regular Ulam sphere sizes were provided, first using Young Tableaux for spheres of any radius centered at $\mathbf{m}_e^r$. Another method used duplicate translocation sets to calculate sphere sizes for a radius of $t = 1$ for any center. Resulting bounds on Code size were also provided. Many open questions remain, including the existence of perfect codes, sphere size calculation methods for more general parameters, and tighter bounds on code size.

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REFERENCES

- [1] A. Barg and A. Mazumdar, "Codes in Permutations and Error Correction for Rank Modulation," in Proc. IEEE ISIT, 2010, pp. 854-858
- [2] I.F. Blake, G. Cohen, and M. Deza, "Coding with permutations, *Inform. Contr.*, vol. 43, pp. 1-19, 1979.
- [3] F. Farnoud, V. Skachek, and O. Milenkovic, "Error-correction in flash memories via codes in the Ulam metric," *IEEE Trans. Inf. Theory*, Vol. 59 No. 5, pp. 3003-3030, 2013.
- [4] F. Farnoud and O. Milenkovic, "Multipermutation Codes in the Ulam Metric," in Proc. IEEE ISIT, 2014, pp. 2754-2758.
- [5] J. Frame, G. Robinson, and R. Thrall, "The hook graphs of the symmetric group," *Canad. J. Math.*, 6, pp. 316-324, 1954.
- [6] W. Fulton, *Young Tableaux*, London Mathematical Society Student Texts. Cambridge University Press, 1997.
- [7] F. Gologlu et al., "New Bounds for Permutation Codes in Ulam Metric," in Proc. IEEE ISIT, 2015, pp. 1726-1730.
- [8] A. Jiang, R. Mateescu, M. Schwartz, J. Bruck, "Rank Modulation for Flash Memories," in Proc. IEEE ISIT, 2008, pp. 1731-1735.
- [9] A. Jiang, M. Schwartz, and J. Bruck, "Correcting Charge-Constrained Errors in the Rank-Modulation Scheme," *IEEE Trans. Inf. Theory*, Vol. 56 No. 5, pp. 2112-2120, 2010.
- [10] J. Kong and M. Hagiwara, "Nonexistence of Perfect Permutation Codes in the Ulam Metric," in Proc. ISITA, 2016, pp. 691-695.
- [11] F. Sala, R. Gabrys, and L. Dolecek, "Deletions in multipermutations," in Proc. IEEE ISIT, 2014, pp. 2769-2773.
- [12] C. Schensted, "Longest increasing and decreasing subsequences," *Canad. J. Math.*, 13, pp. 179-191, 1961.
- [13] D. Slepian, "Permutation modulation," in Proc. IEEE, Vol. 53, 1965, pp. 228-236.
- [14] R. P. Stanley, *Algebraic Combinatorics. Walks, Trees, Tableaux, and More*, Springer Science+Business Media New York, 2013.
- [15] J. Kong and M. Hagiwara, "Multipermutation Ulam Sphere Analysis Toward Characterizing Maximal Code Size," <https://arxiv.org/abs/1701.03896>, 2017.